Activity #1 – Variables

Instructions

- Carefully analyze the given Java program.
- Trace the references and memory changes step-by-step.
- Complete all required tables and questions.
- Show your logical reasoning where necessary.
- Write your answers in a yellow paper (handwritten).
- Take a clear photo or scan of your work.
- Upload your file in Canvas on or before the deadline.
- Submission Format: Section_LastName_FirstName_CodeAnalysisActivity#1.pdf

Please review the code below.

```java
class Student {
    String name;
    int grade;

    Student(String n, int g) {
        name = n;
        grade = g;
    }
}

public class TestStudent {
    public static void main(String[] args) {

        Student s1 = new Student("John", 85);
        Student s2 = new Student("Maria", 90);
        Student s3 = s2;

        s1.grade = 95;
        s3.name = "Ana";

        System.out.println(s1.name + " " + s1.grade);
        System.out.println(s2.name + " " + s2.grade);
        System.out.println(s3.name + " " + s3.grade);
    }
}
```

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BSIT 2-1N **Object Oriented Programming | Midterm Activity #1**

Please answer the following questions:

PART A – VARIABLE IDENTIFICATION TABLE

Variable Name	**Data Type**	**Variable Type**	**Stored In**
name	String	class variable	Heap memory
grade	int	class variable	Heap memory
Student s1, s2, s3	object	Instance variables	Heap memory
n, g	String, int	Method definition	Method area

PART B: Explanation. Please see example below.

Statement	What Happens?	Memory Impact
Student s1 = new Student("John", 85);	s1 instance variable is declared, is an object, “john is assigned to s1.name 85 is to s1.grade	This is stored in heap memory
Student s2 = new Student("Maria", 90);	s2 instance variable is declared, is an object, “maria is assigned to s2.name 85 is to s2.grade	This is stored in heap memory
Student s3 = s2;	s3 instance variable is declared, but the contents is now identical to s2, which is maria 90	This is stored in heap memory. However s2 and s3 share the same reference.
s1.grade = 95;	s1.grade was updated to 95	This is stored in heap memory. The old value is managed by garbage collector
s3.name = "Ana";	s1.grade was updated to 95 but also s2.grade is updated	This is stored in heap memory. The old value is managed by garbage collector

PART C – OUTPUT

John 95
Ana 90
Ana 90